



Bricks

A LEGO® BRICK MODEL OF

Squirtle

A little water-type with a big curl

42 pieces · 20 steps · 3 sections

Unofficial fan-made building instructions

Not physically build-tested. Every part is a real LEGO element and the model was checked on a computer (connections, collisions, every step, balance), but nobody has built it with real bricks yet.



Before you start

An unofficial fan-made manual

This booklet is not a LEGO product. It was generated from the model file by a computer program and checked part by part.

Not physically build-tested

The model and these steps were checked on a computer only: every part/colour is a real LEGO element, all 115 connections between parts were matched in 3D, no two parts overlap, every part can be pushed into place in its step, and the model's balance was estimated. Not tested: how firmly parts hold, part tolerances and how it copes with handling. Take your time and support the model while you work.

The finished model

About 6.6×5.8 cm and 5.8 cm tall, about 36 g (estimated).

Build the flipper feet and small body, then add the rounded shell and head. Three blue elbow pieces curl into the tail.

You will need

- The 42 pieces in 6 colours listed at the back, with their LEGO element IDs and BrickLink numbers.
- A brick separator for pieces that sit too deep for a finger.

Tips

- Sort the pieces by colour before you start.
- Build the sections in order: each sub-assembly is finished before it is used.
- New pieces in a step have a yellow outline; earlier pieces are drawn paler.

The 6 colours

 Medium Azure: 31
 Tan: 4
 Reddish Brown: 2

 White: 2
 Red: 2
 Black: 1

How to read the steps

Parts and step number

7



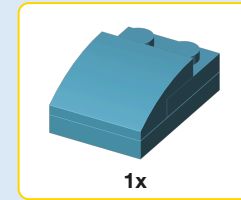
2x

The parts for the step and how many of each, then the step number. The parts of a step are drawn at one scale, so a longer one looks longer; a tile or a long plate also has its size in studs under it.

New parts

The parts added in a step have a yellow outline and are drawn in full colour; the parts from earlier steps are drawn paler.

Sub-assemblies



A yellow-framed picture means: put in the sub-assembly you built in its own section. "Build 4x" means you need four of them.

Sections

Each section starts with its name at the top of the page. Sections are built in the order they appear.

Seen from below

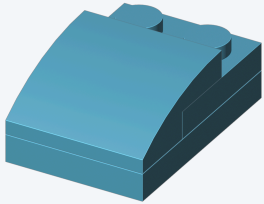
When a step adds parts underneath, the picture shows the model from below. A small label says so (and "seen from behind" likewise). A piece the picture cannot show gets a small picture beside it, ringed in yellow, from the side it can be seen from: where it clicks in.

Insert direction

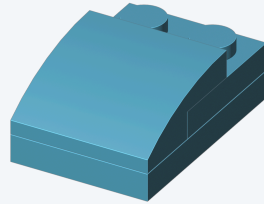
Some parts slide in sideways (axles through bearings, gears onto axles). The picture shows where they end up; push them in along their axle.

Build overview

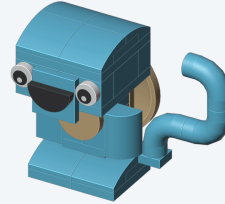
The model is built in 3 sections. Each is finished before it is used.



1. Right foot
steps 1–2



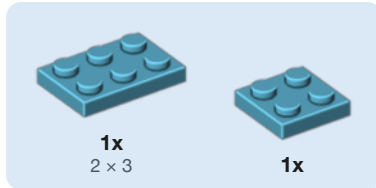
2. Left foot
steps 3–4



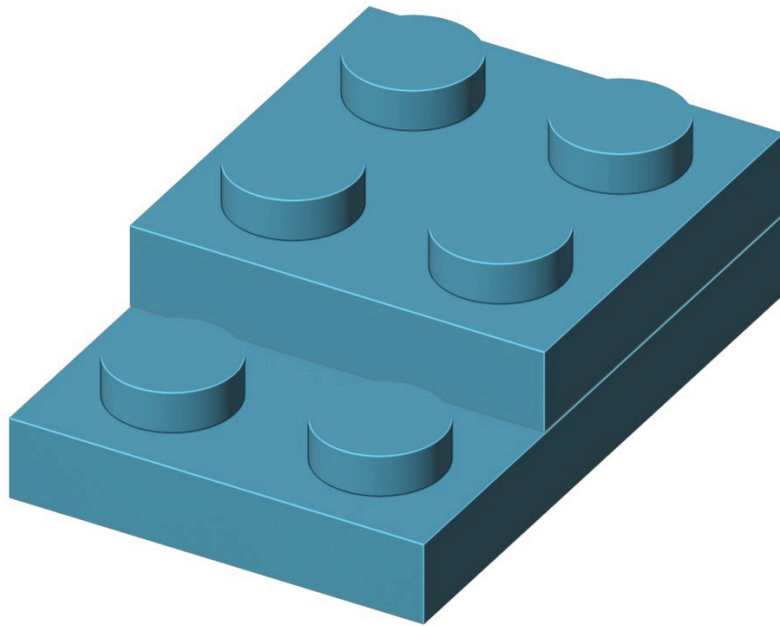
3. Squirtle
steps 5–20

RIGHT FOOT

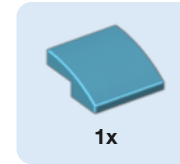
1



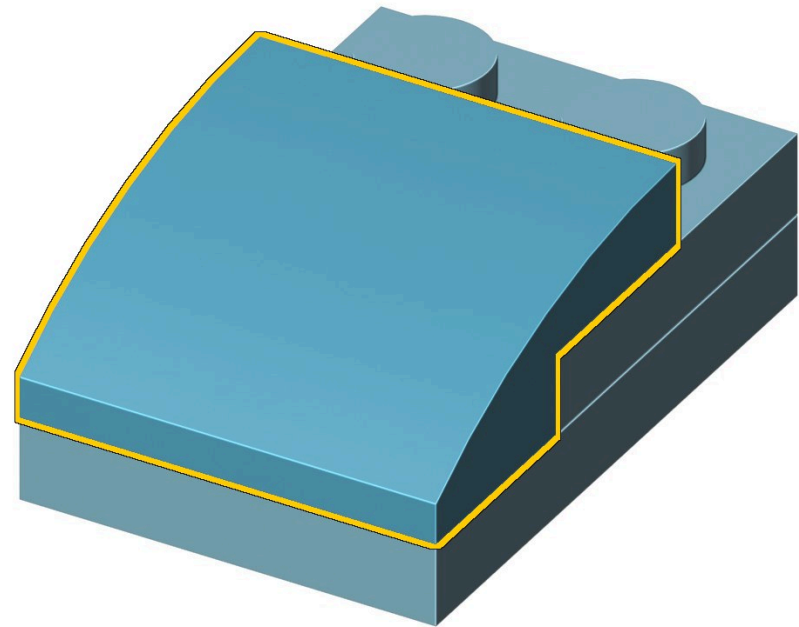
A broad blue sole with a raised heel



2



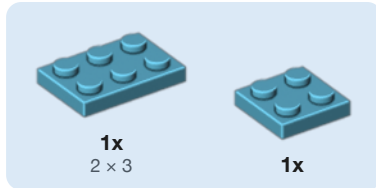
Round the front into a flipper toe



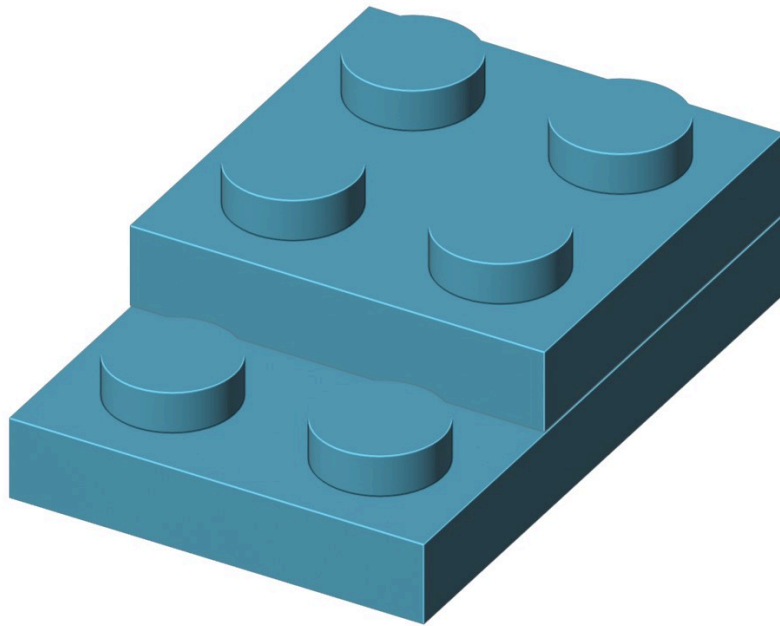
Right foot

LEFT FOOT

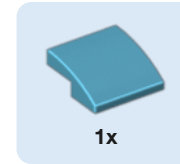
3



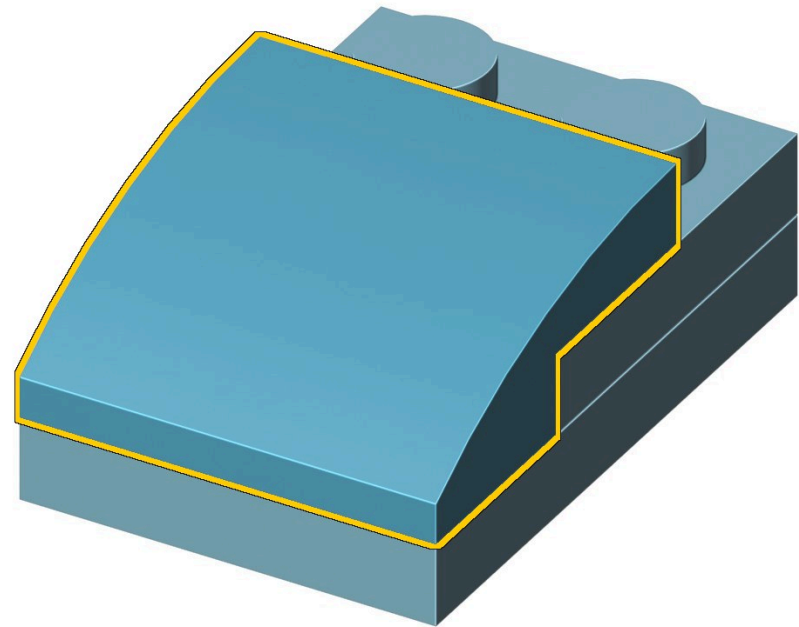
A broad blue sole with a raised heel



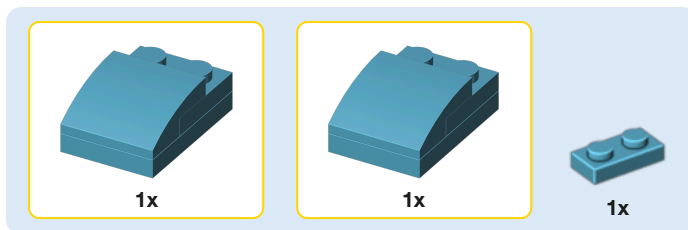
4



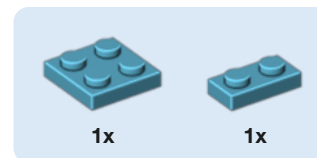
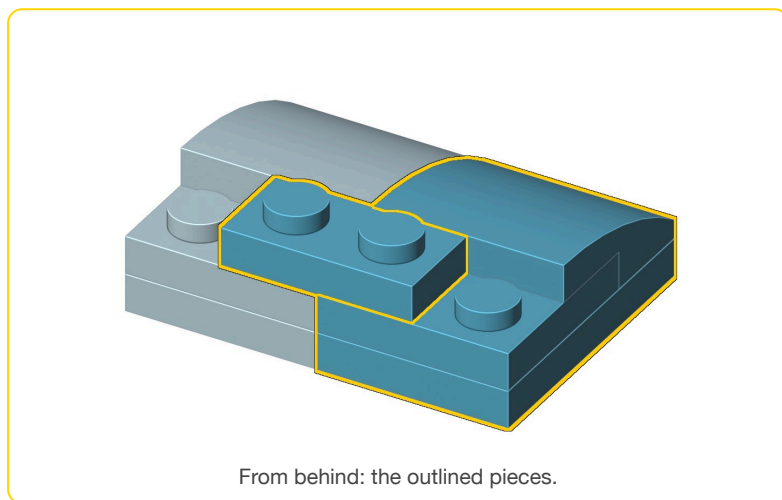
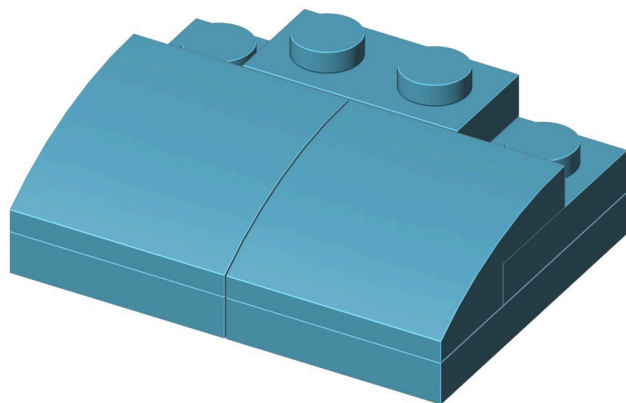
Round the front into a flipper toe



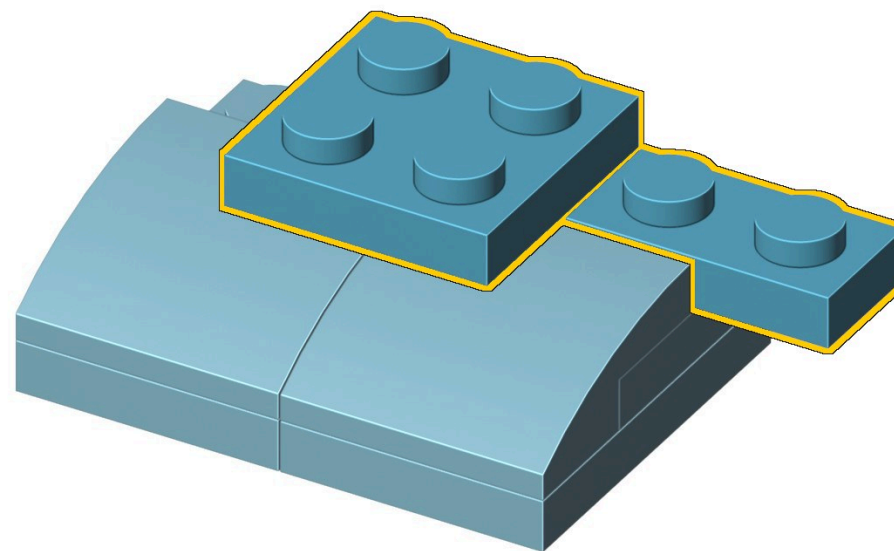
Left foot



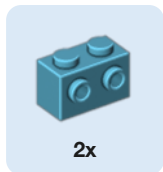
Stand the two flipper feet side by side, toes forward, and tie their heels together with a blue 1 x 2 plate



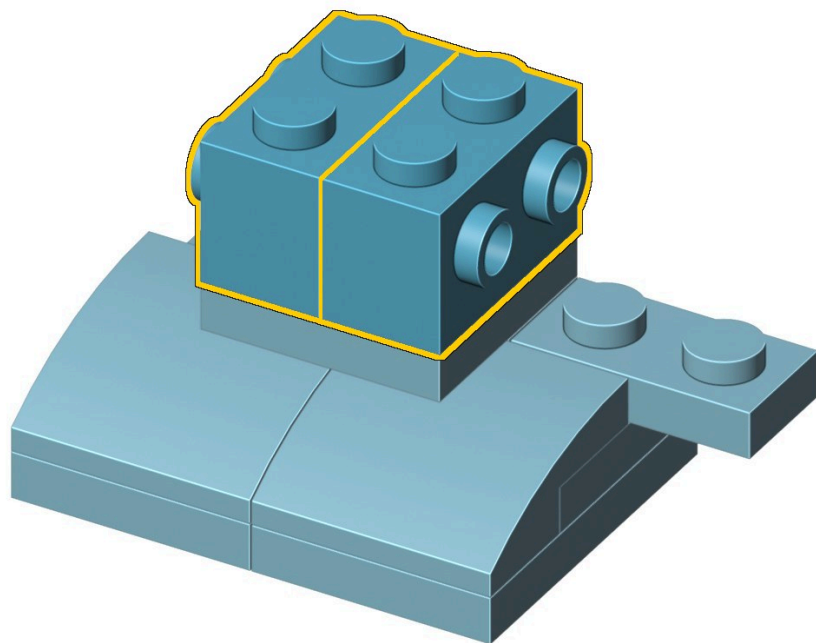
A blue plate behind the right heel starts the outboard tail, and a second hip plate leaves clearance above the curved toes



7

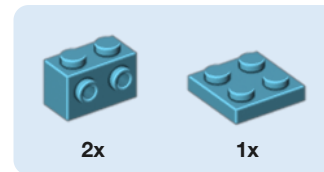


Two bricks turn studs toward the sides for the little arms

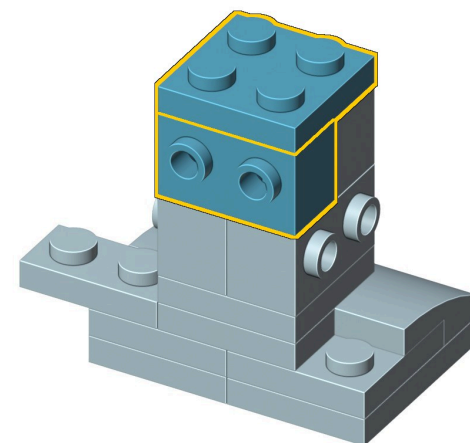
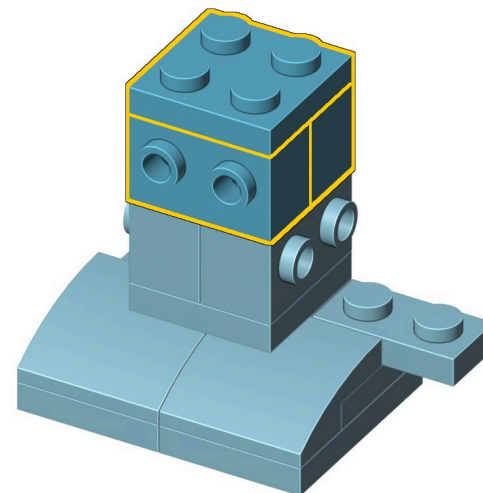


8

Squirtle

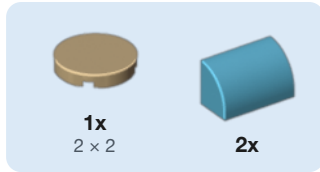


The chest and back each have two studs for the belly and shell

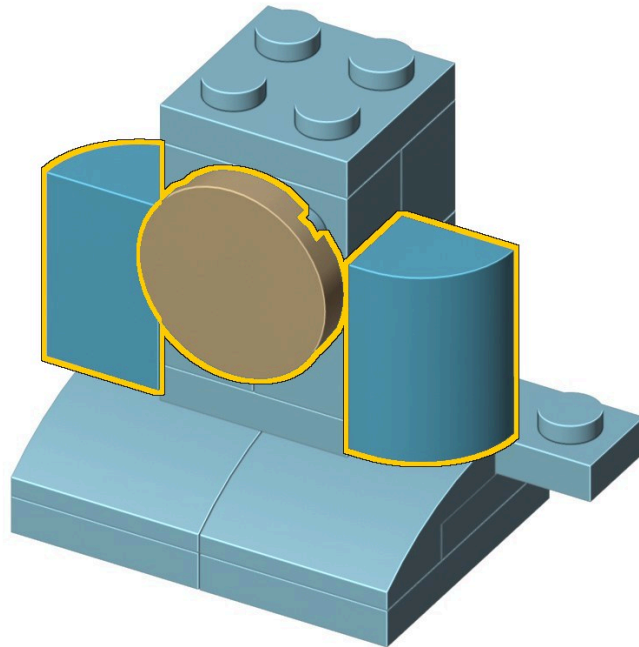


From behind: the outlined pieces.

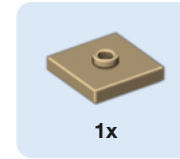
9



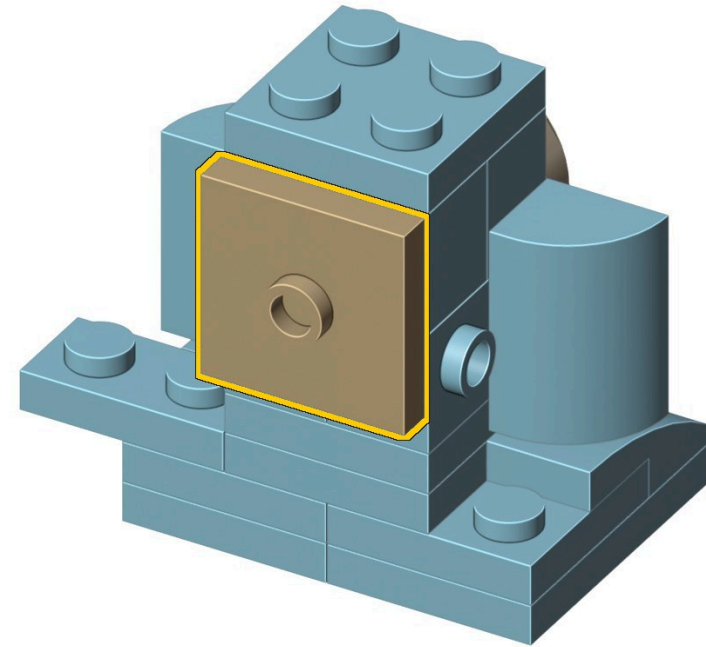
A round tan belly and two rounded blue flippers



10



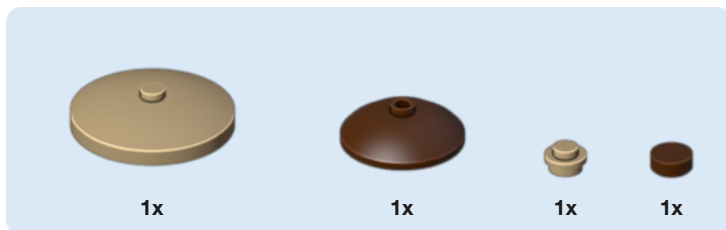
A jumper behind the body centres the shell



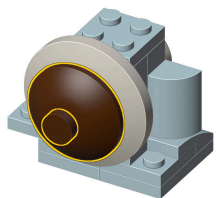
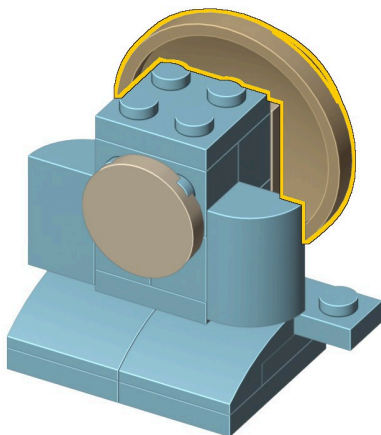
Squirtle

seen from behind

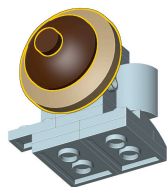
11



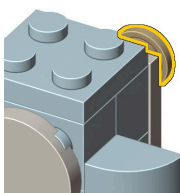
A tan rim around a brown domed shell, finished with a smooth centre



From behind: the outlined pieces.

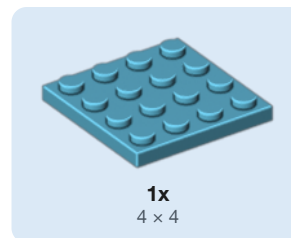


From behind and below: the outlined pieces.

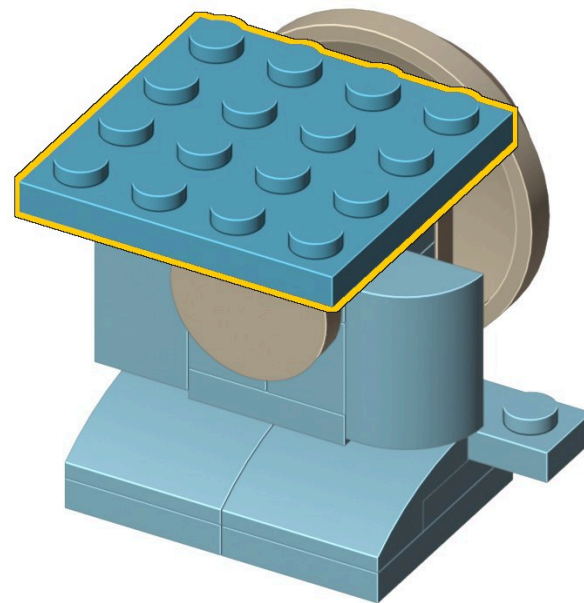


First the outlined piece: the others go on over it.

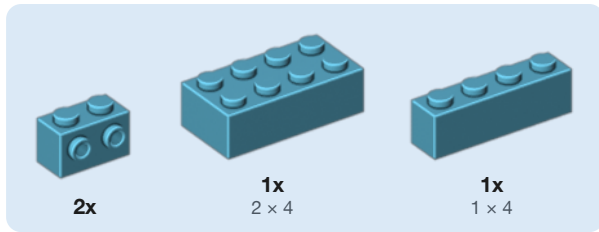
12



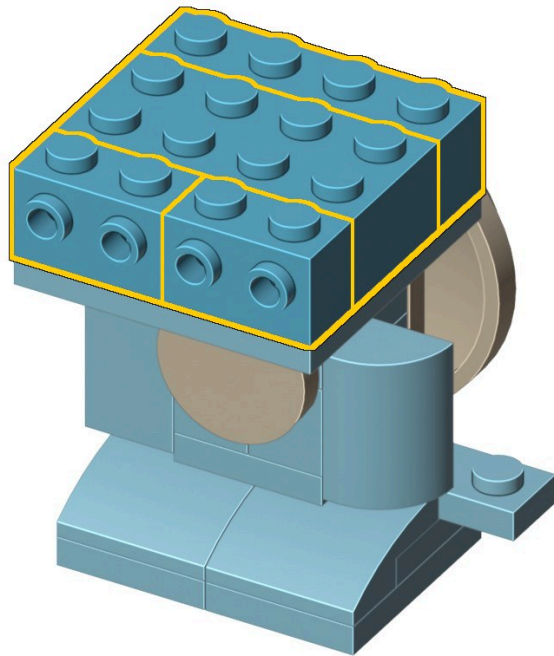
A wide plate supports the oversized head



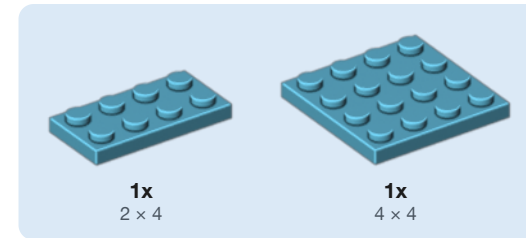
13



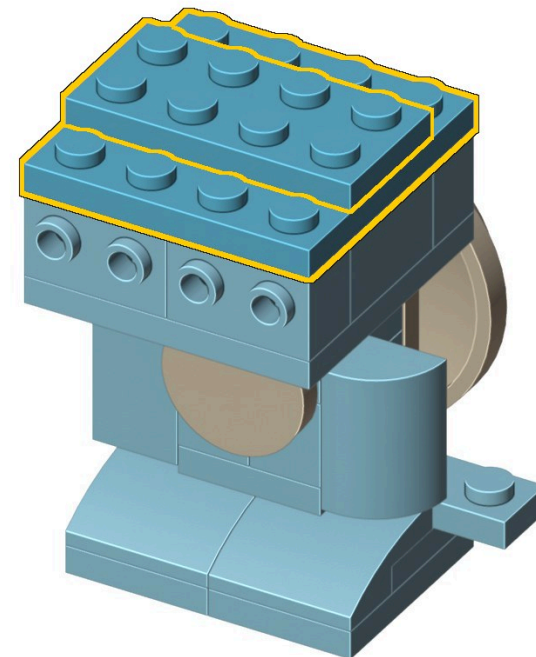
Two eye-support bricks in front and two bricks behind them



14



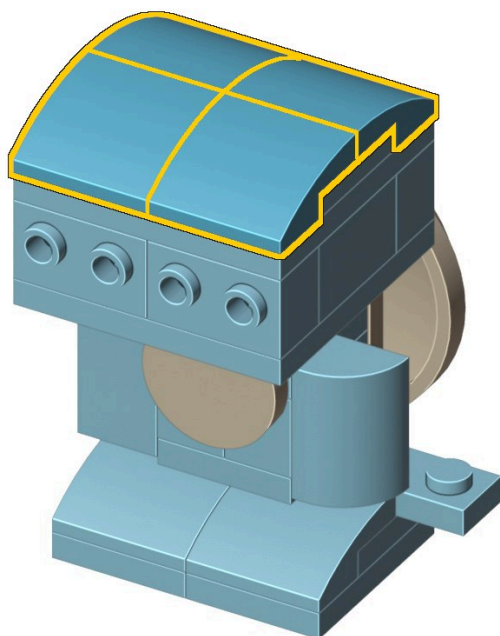
Tie the head together and raise the centre of the crown



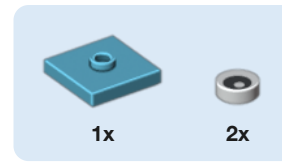
15



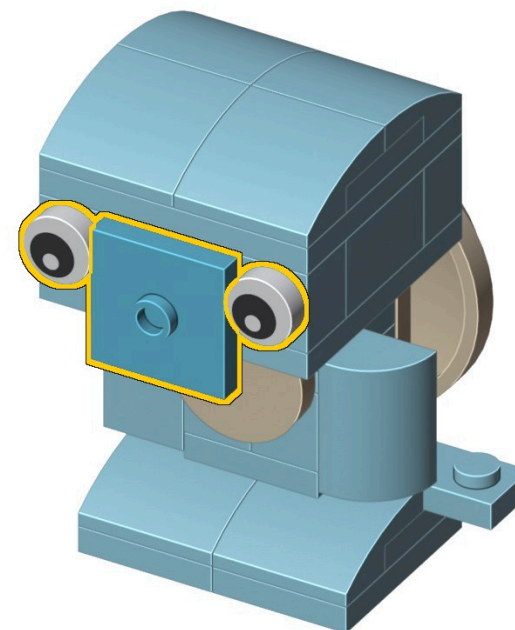
Four curved slopes make a rounded blue crown



16

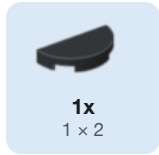


Two printed eyes and a small blue pad for the smile

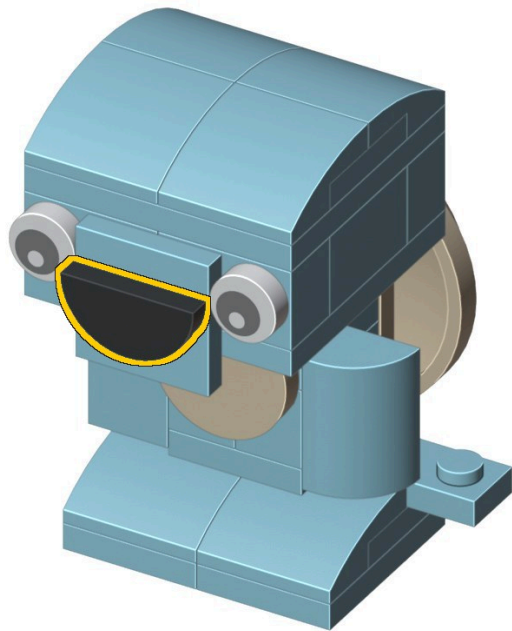


Squirtle

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A little black smile below the eyes

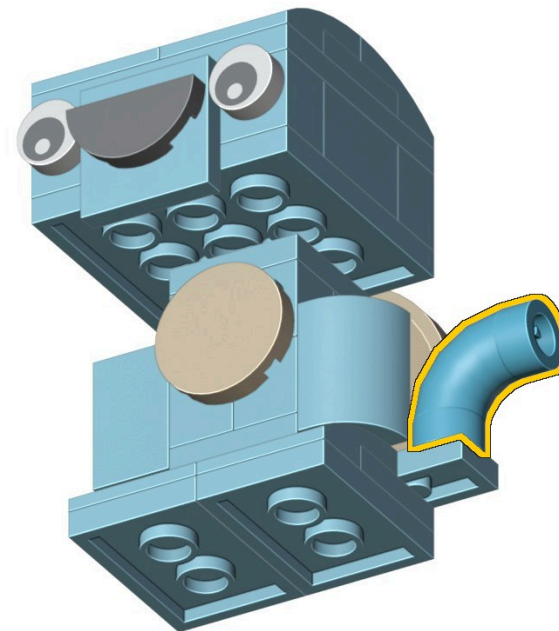


18

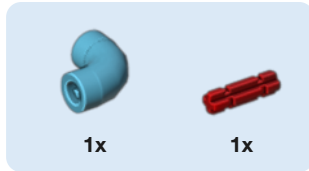


The first blue elbow curls up from the heel

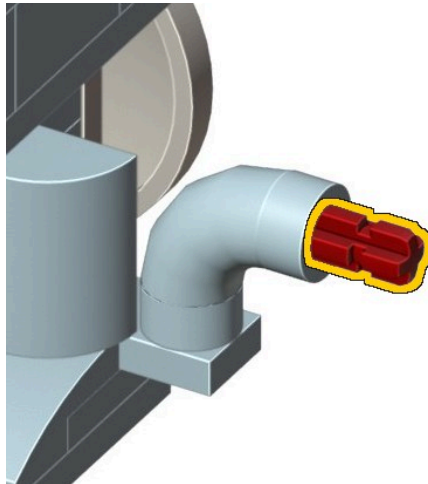
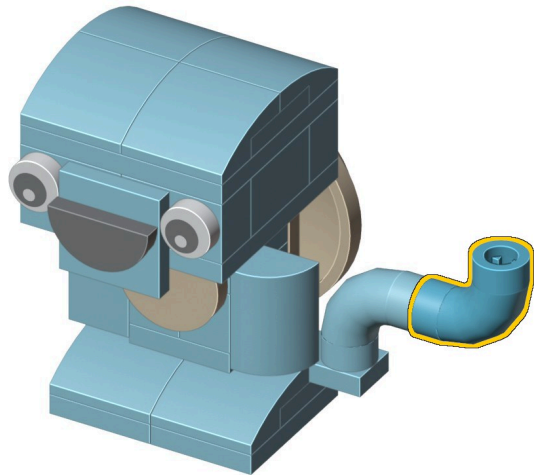
Squirtle
seen from below



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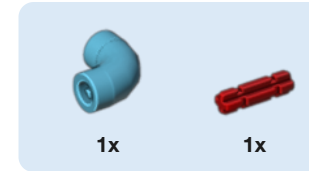


An axle joins the next blue bend, turning the tail upward

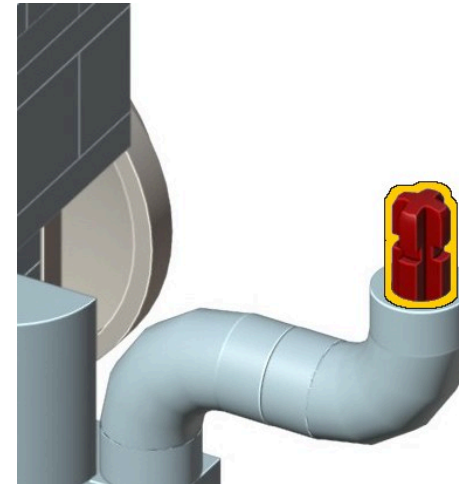
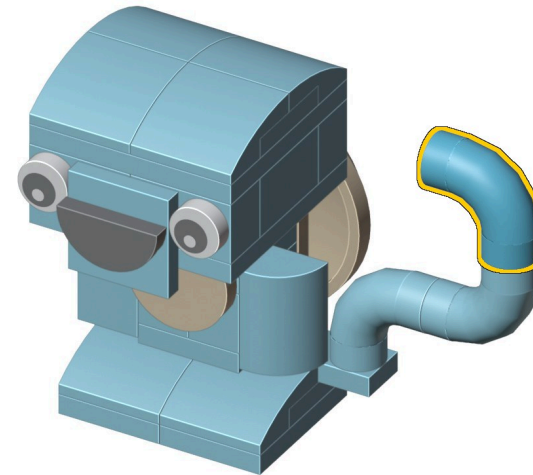


First the outlined piece: the others go on over it.

20



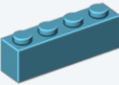
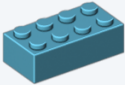
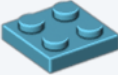
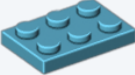
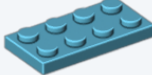
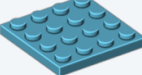
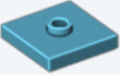
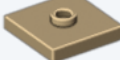
One more axle and elbow curl the tip back toward the head



First the outlined piece: the others go on over it.

Parts inventory

42 pieces in 21 part/colour lines. Bold: the LEGO element ID (what LEGO Pick a Brick sells); below: the colour and the BrickLink part number.

 1x 6510071 Black BrickLink 1748	 1x 6036238 Medium Azure BrickLink 3010	 1x 4625629 Medium Azure BrickLink 3001	 2x 6236578 Medium Azure BrickLink 37352	 6x 6119126 Medium Azure BrickLink 15068	 3x 6369800 Medium Azure BrickLink 25214	 6x 6348142 Medium Azure BrickLink 11211	 2x 6097419 Medium Azure BrickLink 3023
 4x 4625032 Medium Azure BrickLink 3022	 2x 6013828 Medium Azure BrickLink 3021	 1x 6013826 Medium Azure BrickLink 3020	 2x 6141594 Medium Azure BrickLink 3031	 1x 6126090 Medium Azure BrickLink 87580	 2x 4142865 Red BrickLink 32062	 1x 6488491 Reddish Brown BrickLink 43898	 1x 6284586 Reddish Brown BrickLink 98138
 1x 6265305 Tan BrickLink 3960	 1x 4161734 Tan BrickLink 6141	 1x 4565387 Tan BrickLink 87580	 1x 6066344 Tan BrickLink 14769	 2x 6284599 White BrickLink 98138pb007			

About this model

Build the flipper feet and small body, then add the rounded shell and head. Three blue elbow pieces curl into the tail.

Checked on a computer

PASS	real elements	21 part/colour combinations: 0 not real, 0 rare
PASS	connections	115 connections (108 stud, 6 axle, 1 pin); 1 separate piece(s)
PASS	collisions	0 overlapping pair(s) among 42 parts
PASS	buildability	40 insertions tried across 3 submodel(s); 0 problem(s)
PASS	stability	about 36 g; centre of mass inside the base (93% margin); tips at 19.5 deg
PASS	mechanism	no moving parts defined
PASS	electrics	no electrics in this model
PASS	technique	0 note(s)

Not yet built with real bricks

These checks catch most mistakes a designer can make, but not all: clutch power, part tolerances, how stiff a long axle is and how the model survives handling can only be judged with real bricks. If you build it, we would love to hear how it went.

Files

3D model (LDraw, opens in BrickLink Studio, LeoCAD and LDCad), parts lists for BrickLink and LEGO Pick a Brick, and an interactive 3D viewer:

bricks.superfun.games/m/squirtle

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